ELIAS AL HOYEK

Mechatronics & Embedded Systems Engineering Student

Polytech Annecy (MSc) – Seeking a 4–6 month final-year internship, available from February 2027

Contact

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Annecy (74000), France
Open to relocation in France

linkedin.com/in/elias-alhoyek

Portfolio: elias-alhoyek.github.io

Education

2025 – PRESENT

POLYTECH ANNECY – UNIVERSITÉ SAVOIE MONT-BLANC

Master of Science in Mechatronics Engineering

2019 – 2024

AMERICAN UNIVERSITY OF SCIENCE AND TECHNOLOGY, BEIRUT, LEBANON

Bachelor of Science in Computer and Communications Engineering (Biomedical Engineering option)

Technical Skills

- Embedded system design (ESP32, Arduino)
- Real-time (FreeRTOS), BLE & Wi-Fi/UDP
- Signal processing (Python, MATLAB)
- Electromyography (EMG) acquisition
- Sensors & bioinstrumentation
- Circuit analysis & design
- C / C++ programming
- Version control (Git / GitHub)
- Mechanical CAD (Autodesk Inventor)
- Technical project management

Soft Skills

- Complex problem solving
- Rigor and analytical thinking
- Multidisciplinary teamwork
- Communication and teaching
- Leadership and organization
- Adaptability and scientific curiosity

Languages

French	Intermediate B1 (self-assessed)
English	Fluent
Arabic	Native

Profile

Coming from **biomedical engineering**, I moved into mechatronics to get a better handle on the problems people face in healthcare, by designing the devices and machines that solve them. Focused on wearable and medical sensing, I start from the need, define the constraints and validate by measurement, across **physiology, electronics, embedded programming and mechanics**.

Academic & Research Projects

Adaptive Sampling for Low-Power EMG Devices — Ongoing

Research project – Polytech Annecy

- Identified battery life as a key limit of subdermal EMG sensors; designed an amplitude-based adaptive sampling method to extend it.
- Evaluating the trade-off between energy saved and signal fidelity; writing a scientific paper presenting the method and its results.

Fields: signal processing, low-power embedded systems, electromyography (EMG)

Wireless EMG Acquisition System on ESP32 — 2025 – 2026

Master's project – Polytech Annecy

- To keep 8 kHz EMG sampling regular, designed a FreeRTOS architecture (timer interrupt) separating sampling from transmission.
- Prototyped the link in Bluetooth Low Energy (BLE), then switched to Wi-Fi/UDP when BLE could not sustain a reliable 8 kHz stream; validated the chain on the received signal.

Technologies: ESP32, FreeRTOS, BLE, Wi-Fi/UDP, C/C++, EMG sensors

Exoskeleton Device for Muscle Atrophy Rehabilitation — June 2024

Final-year project – AUST

- Designed a device addressing muscle atrophy, assisting grip strength and forearm exercise.
- Integrated control, Arduino boards and instrumentation amplifiers, translating physiological constraints into design choices.

Technologies: Arduino, control engineering, bioinstrumentation, electronics

Also: reverse-engineering a calorimeter's analog control board (ongoing).

Professional Experience

Intern – Biomedical Engineering Department — Oct 2023 – Jan 2024

Bellevue Medical Center – Lebanon

- Diagnosed failures and faults on biomedical equipment, identifying their root causes in a hospital setting, under safety standards.
- Took part in maintenance and repair interventions alongside senior technicians.

Extracurricular Activities

Secretary – IEEE AUST Student Branch · organized events — 2021 – 2023

Mentee – IEEE EMBS Mentoring Program · analyzed data — 2023

Teaching Assistant – AUST Robotics Camp · led workshops — 2023